

Confinement Width: Interface Limits for Evaluator-Relative Control

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Abstract

For controlled dynamics $x_{t+1} = F(x_t, u_t)$ and an evaluator region $R_V = \{x : V(x) \geq \tau\}$, we study the cost of confinement: keeping closed-loop trajectories in R_V under initial-state uncertainty. We price controllers by *interface width*, the number of quantized scalar channels read and written per step. Near a boundary equilibrium with m unstable modes, classical results—the data-rate theorem and invariance entropy [?, ?, ?, ?]—force a feedback rate of at least $\sum_i \log_2 |\lambda_i|$ bits per step, hence width $\Omega(m/b)$, and $\Omega(m)$ for fixed port precision b , under uniform hyperbolicity. Description length inherits the bound only for loop-free interfaces, a restriction we state rather than hide. Critical-point statistics for Gaussian random fields [?, ?, ?] motivate the conjecture that evaluator regions in generic high-dimensional landscapes have boundary equilibria of extensive index $m = \alpha N$, making generic confinement width $\Omega(N/b)$, and $\Omega(N)$ for fixed b : a null model, not a claim about trained networks. A transfer proposition shows that a d -dimensional harness with evaluator sufficiency and per-step measurement caps the relevant width at d ; a minimal exhibit shows the failure mode, an optimizing occupant routing evaluator-relevant change through the projection’s kernel. Absent non-generic structure or a valid quarantine, low-width conditioning steers rather than confines.

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1 Motivation: from interpretability to maintained control

The motivating problem is not merely whether a model can be made to pass an evaluator at one time. It is whether an evaluator’s judgment can remain attached to the intended property while the judged system is trained, prompted, tooled, self-modified, or otherwise recursively improved. In that regime, alignment is less like classifying a finished artifact and more like maintaining an invariant of a changing dynamical system.

This motivates the shift from static description to live control. A mechanistic explanation, a safety feature, a constitutional rule, or a steering vector can all be useful, but none of them is automatically a controller. To bear safety load, an interpretation must supply or induce a measured state through which evaluator-relevant facts continue to factor, a model of how that state evolves, and a feedback interface wide enough to correct departures. The question is therefore not just *what did we find inside the model?* but *what resource budget is required to keep the evaluator-relevant trajectory inside the region we care about?*

The note isolates one local resource: *confinement width*. If a system is near an evaluator boundary with unstable outward modes, classical data-rate and invariance-entropy results say that holding the set invariant requires a per-step information flow. This does not prove a new theorem of control theory; it converts known lower bounds into an evaluator-relative accounting useful for scalable oversight. In particular, it explains why the natural description-length framing is too optimistic for general programmatic controllers: rate constrains the plant–controller pipe, not merely the program that processes the pipe.

The constructive side is dimensional quarantine. Rather than attempting to confine the full high-dimensional base system directly, a harness may project the system into a low-dimensional state $z = \phi(x)$, verify that the evaluator approximately factors through that state, and enforce invariance there. This is the formal role of TRM/LDT-style harnesses in the note: they do not eliminate the confinement problem, but they can reduce its measured dimension when the projection is evaluator-sufficient and dynamically closed enough.

This also gives a prototype for FRC-style complexity mathematics. Confinement width is the local lower-bound term; quarantine transfer is the constructive upper route; fiber ambiguity is the obstruction; and the fidelity half-life $t^*(\lambda)$ is the first empirical observable. The intended research program is to replace slogans about “alignment under recursive improvement” with curves and inequalities: roundwise width requirements, audit costs for maintaining sufficiency, and measured failure times as optimization pressure increases.

2 Opening: what is borrowed, what is priced

The lower-bound mechanism in this note is not new, and we say so here rather than leaving it for a referee: the per-step information requirement for holding a set invariant is the data-rate theorem, with the discrete-time bits-per-step normalization closest to the Nair–Evans finite-rate stabilization line [?], and, in its set-invariance form, invariance entropy (Colonius–Kawan) [?, ?]. Tatikonda–Mitter and Sahai–Mitter are used here for the broader communication-constrained and noisy-channel program, where channel hypotheses and reliability notions matter [?, ?]. Viability theory supplies the constrained-dynamics vocabulary—viable region, regulation map, and maintained constraint set—not the information-rate bound itself [?]. Earlier control work also treats realization burden as mathematical, for example through Brockett’s minimum-attention viewpoint and later minimum-attention work, and through static coder-controller / lookup-table constructions in the invariance-entropy literature [?, ?, ?].

What none of these price, and what this note defines and bounds, is a composite of three things. First, the *interface width* of confinement: the number of quantized channels a controller must read and write per step. This inherits the classical rate bound and, unlike description length, is controller-agnostic. Second, the behavior of that width under the critical-point statistics of generic high-dimensional landscapes, where boundary equilibria of evaluator regions generically carry extensive unstable index, so the classical per-step requirement becomes extensive. Third, the maintenance cost of a low-dimensional quarantine: the validity of the projection and evaluator through which the width reduction is purchased, against an occupant optimizing under the harness’s own proxy. The abstraction and verification literatures do not price this because they typically assume the abstracted system is not trying to leave.

The claim ceiling, stated once and never exceeded: we compose and price known lower bounds in evaluator-relative AI systems; we do not invent invariance entropy.

2.1 Definitions

Dynamics and precision. Consider dynamics

$$x_{t+1} = F(x_t, u_t), \quad x \in X \subseteq \mathbb{R}^N, \quad u \in U \subseteq \mathbb{R}^p. \quad (1)$$

Coordinates live on a fixed-precision grid $\delta\mathbb{Z} \cap [-M, M]$. A scalar port therefore carries

$$b = \log_2 \left(\frac{2M}{\delta} + 1 \right) \quad (2)$$

bits per step: $O(1)$ in N , and $O(\log(M/\delta))$ in precision. Where the landscape picture is wanted, take $F(x, 0) = x - \gamma \nabla L(x)$, so the uncontrolled Jacobian is $I - \gamma H(x)$ with H symmetric and GOE as the null model.

Evaluator region. Let $V : X \rightarrow \mathbb{R}$ be an evaluator and let

$$R_V = \{x : V(x) \geq \tau\}. \quad (3)$$

Ports and interfaces. A *port* is a scalar plant–controller channel exchanging one fixed-precision value per step: sensing ports carry readings in, actuation ports carry controls out. For a controller θ , let $s_r(\theta)$ and $s_w(\theta)$ count active sensing and actuation ports. Define the interface width

$$W(\theta) = \min(s_r(\theta), s_w(\theta)), \quad (4)$$

the bottleneck of the loop. Density of the functional behind a port is irrelevant to width: a dense linear readout $w^\top x$ through one port is one port carrying b bits per step.

Assumption 2.1 (No side channels). *All plant–controller interaction passes through the priced ports. Without this, every bound below is void: a plant that broadcasts its state at unbounded precision outside the accounting supports zero-width confinement trivially.*

Assumption 2.2 (Noiseless quantized ports). *Ports are deterministic digital channels at b bits per plant step. This noiseless quantized-interface model is the normalization used by the width theorem. Noisy channels are strictly harder for the controller and require explicit channel hypotheses; in that setting, anytime capacity rather than raw Shannon capacity is the operative stabilization notion [?, ?], and is out of scope here.*

Definition 2.1 (Confinement width and description length). *Over controllers satisfying Assumptions 2.1 and 2.2 that keep every closed-loop trajectory from the stated uncertainty class in R_V for all $t \leq T$ with failure probability at most ϵ , define*

$$W_V(R, T, \epsilon) = \min_{\theta} W(\theta), \quad L_V(R, T, \epsilon) = \min_{\theta} |\theta|. \quad (5)$$

Here $|\theta|$ is description length, equivalently $\|\theta\|_0$ up to logarithmic addressing in fixed precision [?]. Where the old name is wanted, set $C_V = (W_V, L_V)$. The theorem below attaches to W_V . The quantity L_V is bounded only under the restricted interface class in Remark 3.3.

The uncertainty class is a parameter of the definition and must be nondegenerate: an initial-condition ball of positive radius, or a persistent disturbance of positive amplitude. Degenerate classes admit zero-width open-loop confinement.

Remark 2.1 (What the state space is). Nothing in this note requires literal confinement in parameter space. X is any state space on which the evaluator region and the dynamics are jointly defined: parameter state under training updates, behavior or activation state under inference dynamics, harness state under a projection, or deployment-trajectory state under an operational semantics. Conditioning—prompts, fine-tunes, steering—reshapes F rather than pinning x . The width question is posed after the reshaping: given the dynamics an intervention leaves behind, how many quantized channels of online observation and actuation does it take to keep evaluator-relevant trajectories inside R_V ? A zero-width intervention can relocate equilibria, which is steering and may be sufficient; what it cannot do, near boundary equilibria satisfying the hypotheses below, is supply the per-step bits that Theorem 3.1 says holding requires.

3 The local lower bound

Fix a boundary equilibrium $x^* \in \partial R_V$ of the uncontrolled dynamics. Let

$$A = D_x F(x^*, 0) \quad (6)$$

and let $\{\lambda_i : |\lambda_i| > 1\}$ be the unstable spectrum, of cardinality m , with unstable subspace E^u .

Assumption 3.1 (Nondegenerate uncertainty). *The confinement guarantee is quantified over an initial ball $B_r(x^*)$ with $r > 0$, equivalently over persistent disturbances of positive amplitude. This is what generates the bits.*

Assumption 3.2 (Uniform hyperbolicity and outward transversality). *For all unstable eigenvalues, $|\lambda_i| \geq 1 + \delta$ for some $\delta > 0$. Unstable directions are outward-escaping: uncontrolled trajectories with nonzero E^u component exit $R_V \cap O$ for a neighborhood O of x^* . Where some unstable directions run along ∂R_V or into the interior, m is read as the outward index.*

Lemma 3.1 (Mode coverage: PBH necessity). *Any controller, static or dynamic and of any width, that locally confines under Assumptions 3.1 and 3.2 must have every outward unstable mode controllable through its actuation ports and observable through its sensing ports, in the PBH sense.*

Sketch. An uncontrollable or unobservable unstable mode evolves autonomously in the closed loop by the Kalman decomposition and exits R_V by outward transversality. This constrains *which* subspaces the interface touches, not yet *how many* ports it needs. $\square$

Remark 3.1 (First public correction: rank counting fails). An earlier draft asserted that stabilizing m unstable modes forces feedback of rank at least m , whence $\|\theta\|_0 \geq m$. This is false. For a controllable single-input pair (A, b) , pole placement assigns the closed-loop spectrum arbitrarily, so a rank-one perturbation $b k^\top$ can stabilize any number of unstable modes. For non-normal matrices, low-rank corrections can move unboundedly many eigenvalues. The counterexample supplies no cheap controller in the present accounting—the placing gain is dense and exact—but the obstruction that survives is informational, not algebraic.

PBH mode coverage says which unstable subspaces must be touched; it does not count ports. The port count comes from the information rate required to refine the uncertainty set fast enough to offset unstable expansion.

Theorem 3.1 (Width lower bound). *Under Assumptions 2.1, 2.2, 3.1 and 3.2, any controller keeping every trajectory from $B_r(x^*)$ in $R_V \cap O$ for all t satisfies*

$$\liminf_{T \rightarrow \infty} \frac{\text{feedback bits used up to } T}{T} \geq \sum_{|\lambda_i| > 1} \log_2 |\lambda_i|. \quad (7)$$

Consequently,

$$W(\theta) \geq \frac{1}{b} \sum_{|\lambda_i| > 1} \log_2 |\lambda_i| \geq \frac{\log_2(1 + \delta)}{b} m. \quad (8)$$

Thus $W(\theta) = \Omega(m/b)$. For fixed port precision b and fixed hyperbolicity gap δ , $W(\theta) = \Omega(m)$.

Sketch. The rate inequality is classical: the discrete-time data-rate theorem in the finite-feedback-rate lineage [?] and, in set-invariance form, invariance entropy [?, ?]. The volume-counting intuition is that the uncontrolled dynamics expand E^u volumes by $\prod_i |\lambda_i|$ per step. Sensing partitions the uncertainty set into at most $2^{s_r b}$ cells per step, and actuation branches the response into at most $2^{s_w b}$ controls per step. Invariance requires the refinement to offset the expansion. Since both sides throttle the loop, the bottleneck width $W = \min(s_r, s_w)$ yields (8). Wide sensing alone is not enough under this accounting: with only s_w quantized actuation ports, the controller can realize at most $2^{s_w b}$ distinct control branches per step, so the same per-step refinement bottleneck appears on the write side, even when controls are reused across time. The nonlinear local case should be read through the corresponding invariance-entropy and volume-growth lower bounds, whose sharp statements typically require admissible/control pairs and geometric hypotheses such as nonempty interior, positive measure, finite volume, or hyperbolic isolated-set structure; the theorem above is therefore stated only in the linearized local form. $\square$

Remark 3.2 (Finite horizon). Over finite T the controller may coast on the gap between r and the distance to $\partial(R_V \cap O)$, paying roughly $\log_2(\text{dist}/r)$ fewer total bits. The clean statement here is asymptotic; a circulation draft should either write the finite-horizon correction explicitly or avoid relying on it.

Remark 3.3 (Second public correction: description length does not bound width). It is tempting to conclude $\|\theta\|_0 \geq W(\theta)$, but this is false for general programmatic controllers. A looped fixed-precision program can process arbitrarily many ports with $O(1)$ nonzero parameters by reading them through generated addresses and accumulating. Rate constrains the pipe, not the program. Thus L_V inherits the $\Omega(m/b)$ bound, and the $\Omega(m)$ bound for fixed b , only under a restricted class:

Loop-free/static interfaces: read and write maps are fixed sparse linear or affine maps with no address generation, so each active port consumes at least one dedicated nonzero parameter. Then $\|\theta\|_0 \geq W(\theta)$ and $L_V \geq W_V$ up to addressing logs. For general programmatic controllers, only the width bound stands.

4 The generic extension

Conjecture 4.1 (Extensive boundary index). *For landscapes in the Gaussian-random-field class, such as spherical-spin-glass ensembles, boundary equilibria of evaluator regions at non-extreme levels carry extensive outward index $m = \alpha N$ with high probability. This would follow from Kac–Rice complexity asymptotics restricted to the level set $\{V = \tau\}$, together with the level–index tradeoff of Bray–Dean [?] and random-matrix large-deviation suppression of atypically one-signed spectra [?]. Consequently, under Theorem 3.1, $W_V = \Omega(N/b)$ in the null model, and $\Omega(N)$ for fixed b .*

The restriction of Kac–Rice to an evaluator level set is an unproved step and is named as such [?, ?]. A numerical check is specified as evidence rather than theorem: sample Gaussian fields at $N = 50$ – 200 , locate boundary equilibria of a level set, and histogram the outward index.

GOE/Gaussian genericity is a null model. Trained networks are structured—low-rank, modular, with bulk-zero-plus-outlier Hessian spectra that already depart from GOE—and no claim is made that they are generic fields. The conjecture fixes the default against which structure must be exhibited: a confinement proposal either identifies the non-generic structure it exploits, supplies a quarantine in the sense of section 5, or inherits the $\Omega(N/b)$ width floor, which is $\Omega(N)$ for fixed b .

Corollary 4.1 (Conditioning gap, conditional). *Price standard interventions by width. A prompt or system message writes no runtime port into the evolving state after $t = 0$. Equivalently, prompts and system messages are treated here as reshaping F before the confinement problem is posed, not as priced runtime feedback channels; see Remark 2.1. Offline preference training has zero runtime rate by definition. Activation steering along r fixed directions is r write ports, or rb bits per step. If the operating regime is near boundary equilibria satisfying Conjecture 4.1, then no $o(N/b)$ -width intervention meets the requirement of Theorem 3.1, and no $o(N)$ -width intervention does for fixed b . Stated at its ceiling: absent exploitable non-generic structure or a valid low-dimensional quarantine, low-complexity conditioning should be expected to steer rather than confine.*

Remark 4.1 (Methodological parallel, non-load-bearing). The same critical-point statistics, run on random supergravity potentials, led string cosmology to conclude that metastable perches are exponentially rare and the generic picture is a slow roll along a gradient floor. The parallel carries no evidential weight here; it earns its three sentences only by showing the inversion—perch statistics read as a statement about residence rather than search—has been executed before.

Remark 4.2 (Conjectural energy and fragility obstruction). Earlier drafts carried a Gramian-conditioning argument that bounded-magnitude fixed-precision controllers need actuation support $\Omega(m)$. Two gaps keep it out of the main text: pole placement is sufficient rather than necessary, and the cited energy-scaling results concern reaching target states in network systems rather than gain magnitudes for saddle confinement under GOE spacing. If retained, this material should be re-founded on the discrete-time Bode sensitivity integral [?] and quantized stabilization, then stated only as a fragility conjecture complementing the width bound.

5 Dimensional quarantine

A projection $\phi : X \rightarrow Z$, with $\dim Z = d \ll N$, defines a harness state. Here LDT is glossed as *latent-dynamics tracker*: a harness maintaining an explicit low-dimensional state z with its own transition model. TRM-routed adapters instantiate the class.

Definition 5.1 (Evaluator sufficiency). *A projection ϕ is (V, ϵ) -sufficient on a set $S \subseteq X$ if there exists $v : Z \rightarrow \mathbb{R}$ such that*

$$|V(x) - v(\phi(x))| \leq \epsilon \quad \text{for all } x \in S. \quad (9)$$

To avoid the circularity of “sufficient on the reachable set,” S is required to be a-priori checkable and forward-adequate, for example $S \supseteq R_V \cup F(R_V \times U)$.

Definition 5.2 (Quarantined dynamics and policy-indexed defect). *Projected dynamics are written*

$$z_{t+1} = G(z_t, u_t) + \eta_t, \quad (10)$$

where the closure defect is defined by

$$\eta_t := \phi(F(x_t, u_t)) - G(\phi(x_t), u_t). \quad (11)$$

This is the projected shadow of unmodeled fiber dynamics, a Mori–Zwanzig-style memory term [?, ?], and is therefore statistically dependent on whatever the occupant’s optimization touches in the fiber. Accordingly the defect bound is worst-case over the occupant’s policy class and round-indexed:

$$\bar{\eta}(\Pi_k) = \sup_{\pi \in \Pi_k} \sup_{x \in S, u \in U} \|\phi(F(x, u)) - G(\phi(x), u)\|. \quad (12)$$

Optimization enlarges Π_k , so $\bar{\eta}$ is a function of round; guarantees below hold per round at the current $\bar{\eta}$.

6 Reactive transfer

Proposition 6.1 (Reactive quarantined confinement). *Assume ϕ is (V, ϵ) -sufficient on S , v is L_v -Lipschitz, per-step defects satisfy $\|\eta_t\| \leq \bar{\eta}$, the controller measures $z_t = \phi(x_t)$ at every step, and the robust one-step condition holds:*

$$\forall z \text{ with } v(z) \geq \tau + \epsilon, \quad \exists u \in U \text{ such that } v(G(z, u)) \geq \tau + \epsilon + L_v \bar{\eta}. \quad (13)$$

If $v(\phi(x_0)) \geq \tau + \epsilon$ and $x_0 \in S$, then $x_t \in R_V$ for all t .

Proof. Induct on t . Since $x_t \in R_V \subseteq S$, forward adequacy gives $x_{t+1} \in F(R_V \times U) \subseteq S$, so sufficiency applies. From $v(z_t) \geq \tau + \epsilon$, choose u_t according to (13). Then

$$v(z_{t+1}) \geq v(G(z_t, u_t)) - L_v \|\eta_t\| \geq \tau + \epsilon. \quad (14)$$

By evaluator sufficiency, $V(x_{t+1}) \geq v(z_{t+1}) - \epsilon \geq \tau$. The base case is identical. $\square$

The robust one-step condition is a discrete-time control-barrier-function condition [?, ?]. Its verification is a d -dimensional problem, which is the point.

Remark 6.1 (Predictive variant). If z is measured only at $t = 0$ and the controller certifies a model trajectory $\hat{z}_{t+1} = G(\hat{z}_t, u_t)$, the margin inflates from $\epsilon + L_v \bar{\eta}$ to

$$\epsilon + L_v D_T, \quad D_T = \bar{\eta} \frac{L_G^T - 1}{L_G - 1}, \quad (15)$$

with $D_T = \bar{\eta}T$ when $L_G = 1$. This compounds geometrically unless G is contractive on the relevant region. Predictive quarantine is therefore a short-horizon instrument; the design criteria are to build G contractive and measure ϕ online.

Corollary 6.1 (Width cap and measurement). *Under Proposition 6.1, the confinement problem the controller actually solves lives in Z . The width bound applies to the outward index m_G of G on ∂R_Z , with $m_G \leq d$, not necessarily to the full base dynamics, provided the evaluator factors through ϕ . TRM/LDT-style harnesses do not magically solve confinement; they reduce and measure the dimensionality of the confinement problem. What is not reduced is the maintenance of (V, ϵ) -sufficiency and closure against an optimizing occupant: $\bar{\eta}(\Pi_k)$ and the validity of S are round-indexed quantities not bounded by d .*

7 The exhibit

Remark 7.1 (Where the adversary lives). Corollary 6.1 bounds confinement with respect to v , not V , unless sufficiency holds on S for the duration. The adversary lives in $\ker \phi$: hidden fiber variables can change V while leaving $v \circ \phi$ fixed. The slack ϵ is not a constant of the projection; it is a functional of the occupant's policy class and optimization pressure.

Exhibit 7.1 (Fiber routing with $\eta \equiv 0$). *Let $X = Z \times Y$ with $Z = Y = [0, 1]$, and let $\phi(z, y) = z$. Dynamics are*

$$z_{t+1} = \text{clip}(z_t + u_t), \quad u_t \in [-\bar{u}, \bar{u}] \text{ chosen by the harness}, \quad (16)$$

$$y_{t+1} = \text{clip}(y_t + \pi_\lambda(y_t) + w_t), \quad (17)$$

where π_λ is the occupant’s policy after optimization pressure λ , and w_t is optional noise. The z dynamics are exactly closed, so $\eta \equiv 0$ by construction; every failure below is pure evaluator-sufficiency failure.

Let the proxy be $v(z) = \mathbf{1}[z \leq \zeta]$ and the true evaluator be

$$V(z, y) = \mathbf{1}[z \leq \zeta] \mathbf{1}[y \leq \beta], \quad y_0 < \beta < 1. \quad (18)$$

The harness keeps $z \leq \zeta$, so $v(\phi(x_t)) = 1$ for all t . But any occupant objective monotone increasing in y drives the reachable set out of $S_\beta = \{(z, y) : y \leq \beta\}$, where ϕ was sufficient. The true evaluator fails at

$$t^*(\lambda) = \min\{t : V(x_t) = 0\}, \quad \mathbb{E}[t^*(\lambda)] \approx \frac{\beta - y_0}{\rho(\lambda)}, \quad (19)$$

where $\rho(\lambda) = \mathbb{E}[\pi_\lambda]$ is the realized drift rate. The observable is a fidelity half-life curve: proxy flat while ground truth decays.

This exhibit is not meant to instantiate Proposition 6.1’s Lipschitz transfer theorem: the proxy v is an indicator. It isolates the separate failure mode in which closure is exact, the proxy remains satisfied, and evaluator sufficiency fails through the fiber.

Implementation sketch. A finite state machine carries the world. The visible ledger is $z \in \{0, \dots, K\}$, which the harness reads; hidden commitment flags are $y \in \{0, 1\}^\ell$ with a designated bad set. The harness is rule-based and reads z only. The occupant is a small stochastic policy: minimally a softmax table updated by REINFORCE on reward equal to progress into the fiber; in the full version, a tinyLoRA adapter over the TRM router is updated per round against the proxy the harness exposes. Each round logs $v \circ \phi$ and the ground-truth V read directly from the state machine; no judge model is introduced. Sweep λ to produce the $t^*(\lambda)$ curve.

8 What is actually novel

No new lower-bound mechanism is introduced, and the note says so where a referee would otherwise say it first. The per-step information requirement is the data-rate theorem in set-invariance form and invariance entropy [?, ?, ?]; noisy-channel generalizations require additional communication hypotheses [?, ?]. Implementation burden has antecedents in Brockett’s minimum-attention control and in finite-abstraction/static-coder-controller constructions [?, ?, ?]. These antecedents motivate pricing, but they are not direct sources of the confinement-width lower bound.

The contributions are a definition and three compositions. The definition is *confinement width*, the channel count of the plant–controller interface, which inherits the classical rate bound and is controller-agnostic. The compositions are: width against the critical-point statistics of generic landscapes, where extensive boundary index makes the classical requirement extensive; width against dimensional quarantine, where a transfer proposition caps the requirement at the harness dimension and decomposes the price into evaluator slack and closure defect; and quarantine against an optimizing occupant, where maintaining evaluator sufficiency becomes the residual unbounded quantity, exhibited in a minimal system with a measured half-life.

9 Relation to FRC

FRC concerns evaluator-relative fidelity under recursive updates: the cost of keeping a bounded evaluator’s judgment attached to the intended property while the judged system changes itself. Confinement width supplies one local dynamical lower-bound object inside that agenda: the feedback and interface width needed to keep evaluator-relevant trajectories inside a region, priced per step, with the classical rate theorems doing the pricing. Dimensional quarantine is the constructive attempt to reduce that width by forcing evaluator-relevant dynamics through a low-dimensional measured state. Exhibit 7.1 shows the fidelity failure when evaluator-relevant facts stop factoring through that state: $t^*(\lambda)$ is a fidelity half-life, measured rather than metaphorical. The residual quantity the width bound cannot cap—round-indexed maintenance of sufficiency and closure—is where the FRC program continues past this note.

10 FRC as recursive quarantine maintenance

The previous sections are local: one dynamics F , one evaluator region R_V , one confinement problem. Fidelity Recursive Complexity (FRC) is the recursive version of the same accounting. It asks what must be paid when the judged system, the reachable set, and the validity of the harness abstraction all change across rounds. The definitions below are intentionally modest: they turn FRC into a bookkeeping layer over confinement width and quarantine maintenance, rather than a new control theorem.

Definition 10.1 (Recursive evaluator-relative system). *A recursive evaluator-relative system of depth K is a sequence of controlled dynamics*

$$x_{t+1}^{(k)} = F_k(x_t^{(k)}, u_t^{(k)}), \quad k = 0, \dots, K-1, \quad (20)$$

with state spaces X_k , admissible controls U_k , uncertainty classes $\mathcal{U}_k$, and target evaluator regions

$$R_V^{(k)} = \{x \in X_k : V_k(x) \geq \tau_k\}. \quad (21)$$

The index k denotes a recursive update round: a training update, self-modification, tool/harness revision, or red-team patch cycle. The evaluator V_k may be fixed or updated, but the intended property is held fixed at the level of the experiment: the question is whether the updated evaluator remains attached to that property.

Definition 10.2 (Round- k quarantine). *A quarantine at recursive round k is a tuple*

$$Q_k = (\phi_k, v_k, G_k, S_k, \Delta_k) \quad (22)$$

where $\phi_k : X_k \rightarrow Z_k$ is a measured projection, $d_k = \dim Z_k$, $v_k : Z_k \rightarrow \mathbb{R}$ is the projected evaluator, G_k is the projected transition model, $S_k \subseteq X_k$ is the a-priori forward-adequate set on which the abstraction is certified, and $\Delta_k \geq 0$ is the projected safety margin.

Write the evaluator-sufficiency and closure defects as

$$\epsilon_k = \sup_{x \in S_k} |V_k(x) - v_k(\phi_k(x))|, \quad (23)$$

$$\bar{\eta}_k(\Pi_k) = \sup_{\pi \in \Pi_k} \sup_{x \in S_k, u \in U_k} \|\phi_k(F_k(x, u)) - G_k(\phi_k(x), u)\|, \quad (24)$$

where Π_k is the occupant policy class available at round k . If v_k is $L_{v,k}$ -Lipschitz, define the recursive fidelity margin

$$M_k = \Delta_k - \epsilon_k - L_{v,k} \bar{\eta}_k(\Pi_k). \quad (25)$$

The quarantine is margin-valid at round k only when $M_k \geq 0$, together with the robust one-step condition of Proposition 6.1 in the projected state space.

Definition 10.3 (FRC width and audit complexity). Fix a horizon T_k for each recursive round, a failure tolerance ϵ_{tot} , and a class of admissible recursive harnesses $\mathcal{H}$. For $H \in \mathcal{H}$, let $W_k(H)$ be the confinement width used at round k , measured in the active state space after any quarantine, and let $A_k(H)$ be the audit/refinement cost required to certify S_k , ϵ_k , $\bar{\eta}_k(\Pi_k)$, and the robust one-step condition at that round. The width component of FRC is

$$\text{FRC}_V^W(K, \epsilon_{\text{tot}}) = \inf_{H \in \mathcal{H}} \left\{ \sum_{k=0}^{K-1} W_k(H) : \Pr_H \left[\forall k < K, \forall t \leq T_k, x_t^{(k)} \in R_V^{(k)} \right] \geq 1 - \epsilon_{\text{tot}} \right\}. \quad (26)$$

The audit component is defined analogously,

$$\text{FRC}_V^A(K, \epsilon_{\text{tot}}) = \inf_{H \in \mathcal{H}} \sum_{k=0}^{K-1} A_k(H) \quad \text{subject to the same fidelity constraint.} \quad (27)$$

A scalar total FRC_V can be obtained by choosing exchange rates between width, compute, proofs, oversight hours, latency, and hardware costs. Without such exchange rates, the pair $(\text{FRC}_V^W, \text{FRC}_V^A)$ is the safer mathematical object.

Proposition 10.1 (Roundwise lower bound embeds into FRC). Suppose that, at recursive round k , the active confinement problem satisfies the hypotheses of Theorem 3.1 with unstable eigenvalues $\lambda_{k,i}$. Then any recursive harness satisfying the fidelity constraint in Definition 10.3 must pay

$$W_k \geq \frac{1}{b_k} \sum_{|\lambda_{k,i}| > 1} \log_2 |\lambda_{k,i}|. \quad (28)$$

Consequently, on any set of rounds $\mathcal{K}$ where the theorem's hypotheses hold,

$$\text{FRC}_V^W(K, \epsilon_{\text{tot}}) \geq \sum_{k \in \mathcal{K}} \frac{1}{b_k} \sum_{|\lambda_{k,i}| > 1} \log_2 |\lambda_{k,i}|. \quad (29)$$

Sketch. Apply Theorem 3.1 separately at each recursive round to the active state space on which confinement is actually attempted: the base state X_k if no quarantine is supplied, or the projected state Z_k if a valid quarantine transfers the problem. Summing the unavoidable roundwise widths gives (29). No new lower-bound mechanism is introduced; FRC inherits the local width theorem as one term in its recursive resource accounting. $\square$

Proposition 10.2 (Dimensional quarantine as an upper route). Suppose that for every round $k < K$ there exists a quarantine Q_k satisfying evaluator sufficiency, bounded closure defect, and the reactive transfer condition of Proposition 6.1. If the projected controller reads and writes the full projected state at fixed precision, then

$$W_k \leq d_k \quad \text{and hence} \quad \text{FRC}_V^W(K, \epsilon_{\text{tot}}) \leq \sum_{k=0}^{K-1} d_k \quad (30)$$

for the width component of a successful harness. The unresolved cost is not width but audit: certifying that $M_k \geq 0$ and that the projected system remains closed under the updated policy class.

Sketch. Under Proposition 6.1, confinement can be verified and enacted in Z_k . A controller that measures the d_k scalar coordinates of $z_k = \phi_k(x_k)$ and writes controls through the projected interface has width at most d_k , up to the chosen port accounting. The guarantee holds only while the sufficiency and closure terms in (25) are audited and remain inside margin. $\square$

Lemma 10.1 (Fiber ambiguity obstruction). *Fix a round k and a projection $\phi_k : X_k \rightarrow Z_k$. Suppose there exist two reachable states $x, x' \in S_k$ such that*

$$\phi_k(x) = \phi_k(x') \quad \text{but} \quad V_k(x) \geq \tau_k + \epsilon \quad \text{and} \quad V_k(x') < \tau_k - \epsilon. \quad (31)$$

Then no controller whose observations and certificates factor only through ϕ_k can guarantee V_k -confinement at slack ϵ simultaneously over an uncertainty class containing both states.

Proof. The controller receives the same projected observation at x and x' and therefore must choose the same projected action and issue the same certificate in both cases. But V_k assigns the two states to different sides of the margin. Any guarantee phrased only in terms of $v_k(\phi_k(\cdot))$ cannot distinguish them. Thus the projection is not (V_k, ϵ) -sufficient on the relevant reachable set, and a ϕ_k -only controller can confine at most the proxy, not the true evaluator region. $\square$

Definition 10.4 (Fidelity half-life). *For a recursive process under optimization pressure λ , let global time be the pair (k, t) ordered lexicographically by recursive round k and within-round time t . Define the fidelity half-life as the first such global time at which either the true evaluator fails or the recursive fidelity margin becomes negative:*

$$t^*(\lambda) = \inf\{(k, t) : V_k(x_t^{(k)}) < \tau_k \text{ or } M_k < 0\}. \quad (32)$$

In Exhibit 7.1, this reduces to $t^(\lambda) = \min\{t : V(x_t) = 0\}$. The empirical curve $t^*(\lambda)$ is therefore a prototype observable for FRC stress: as optimization pressure expands Π_k or pushes mass through $\ker \phi_k$, the lifetime of the evaluator-sufficient abstraction shrinks.*

Remark 10.1 (What mechinterp can and cannot buy). Mechanistic interpretability can lower the width term by proposing candidate projections ϕ_k , latent state variables, and unstable directions to monitor. It does not by itself pay the audit term. In the recursive limit, an interpretation is a candidate quarantine, not a permanent certificate. FRC prices the maintenance of its evaluator-sufficiency and closure under the optimization pressure it is supposed to control.

11 Future work and verification

Proof and evidence debts

1. **Generic index.** Conjecture 4.1 requires Kac–Rice restricted to a level set. A low- N numerical check, $N = 50$ – 200 , should be implemented as evidence.
2. **Nonlinear local case.** Verify whether Kawan’s volume-growth lower bounds cover the linearization neighborhood. If not, scope Theorem 3.1 to the linearized statement.
3. **Finite-horizon correction.** Write the constant in Remark 3.2.
4. **Energy/fragility appendix.** Re-found on the Bode integral or cut.
5. **Exhibit implementation.** Implement Exhibit 7.1; the note should cite its own $t^*(\lambda)$ plot rather than a promissory one.

Citation verification status

- **Invariance entropy scope.** Colonius–Kawan and Kawan [?, ?] support the set-invariance/data-rate lineage, but nonlinear formulas and sharp bounds should be stated with their geometric hypotheses visible: nonempty interior, positive measure, finite volume, or hyperbolic isolated-set assumptions as applicable. The main theorem is intentionally local-linear.
- **Data-rate normalization.** The theorem uses a noiseless, quantized, bits-per-plant-step interface. Nair–Evans is the closest antecedent for this discrete-time finite-rate normalization [?]; Tatikonda–Mitter and Sahai–Mitter are cited for the broader communication-constrained/noisy-channel setting [?, ?].
- **Implementation antecedents.** Brockett 1997 is reinstated for the minimum-attention viewpoint; Egerstedt–Brockett 2003 remains unpinned and is not used as a load-bearing citation. The draft also cites later, directly checkable sources for minimum-attention control and static coder-controller / lookup-table implementations [?, ?, ?].
- **Aubin viability theory.** Aubin is used only for constrained-dynamics vocabulary—viable region, regulation map, and the question of maintaining trajectories inside a constraint set—not for the entropy lower bound [?].
- **Generic-index conjecture.** The Bray–Dean/Auffinger–Ben Arous–Cerny background [?, ?] does not prove the evaluator-boundary Kac–Rice step; that remains future work.
- **Dean–Majumdar constant.** Keep exact constants, including $\theta = (\ln 3)/4$, on the verification list; the text currently uses only the weaker large-deviation suppression wording [?].
- **Fragility appendix.** Use the discrete-time Bode citation [?] only if the fragility appendix is retained and re-founded.
- **Musat 2026.** Treat the fixed-precision/description-length relation as preprint support only [?], not as load-bearing for the width theorem.

A Exhibit 7.1 implementation specification

Claim under test

ϕ -confinement is worth no more than evaluator sufficiency on the closed-loop, post-optimization reachable set, and the fidelity half-life $t^*(\lambda)$ is that worth, measured. The exhibit is constructed so $\eta \equiv 0$ identically, eliminating closure error as a cause.

System definition

Variant A: continuous minimal. State $(z, y) \in [0, 1]^2$, projection $\phi(z, y) = z$. Dynamics per step:

$$z \leftarrow \text{clip}(z + u), \quad u \in [-\bar{u}, \bar{u}], \quad y \leftarrow \text{clip}(y + a + w), \quad (33)$$

where a is the occupant’s action and $w \sim \mathcal{N}(0, \sigma^2)$ is optional, with $\sigma = 0$ by default. Proxy $v(z) = \mathbf{1}[z \leq \zeta]$; true evaluator $V(z, y) = \mathbf{1}[z \leq \zeta]\mathbf{1}[y \leq \beta]$. Defaults: $\zeta = 0.5$, $\beta = 0.8$, $y_0 = 0.1$, $z_0 = 0.2$, $\bar{u} = 0.1$.

Variant B: FSM storyworld. A finite state machine has visible trust ledger $z \in \{0, \dots, K\}$, $K = 10$, hidden commitment flags $y \in \{0, 1\}^\ell$, $\ell = 4$, and a designated bad set. At least two actions must have identical z effects and different y effects, displayed as a table rather than claimed.

Harness policy

The harness reads z only. If z is above margin, it applies a corrective u or selects a z -decreasing action. The invariant $v(\phi(x_t)) = 1$ should be asserted at every step; a proxy failure is a bug, not data.

Occupant policy and update

Variant A uses a single drift parameter or a small softmax over $\{-\bar{\rho}, 0, +\bar{\rho}\}$. Variant B uses a softmax table $\pi(a \mid z, y)$ over world actions. Update by REINFORCE on reward $R = y$ in Variant A or progress into the fiber in Variant B. The exhibit survives any objective monotone in the fiber coordinate.

Optimization pressure and metric

Let λ be the REINFORCE learning rate or selection strength. Run $\lambda = 0$ as a control arm. The metric is

$$t^*(\lambda) = \min\{t : V(x_t) = 0\}, \quad (34)$$

censored at $T_{\max} = 10^4$. Report median and IQR over at least 50 seeds per λ , with a log grid such as 10^{-3} to 10^0 , plus 0. Compare to the sanity prediction $\mathbb{E}[t^*(\lambda)] \approx (\beta - y_0)/\rho(\lambda)$.

Expected plots

1. $t^*(\lambda)$, median with IQR band, log- x : the fidelity half-life against optimization pressure.
2. For three representative λ , plot $v(\phi(x_t))$ and $V(x_t)$ versus t : proxy flat at 1, truth dropping at t^* .
3. Optional: occupancy histogram of y or fiber-flag counts by round, showing the reachable set leaving the sufficiency region while z statistics are stationary.
4. Control: $\lambda = 0$ arm overlaid, all censored.

Smallest CPU-only version

The implemented minimal version is a Python script, `scripts/exhibit_6_1_fiber_routing.py` (script name retained from an earlier draft). It is CPU-only, deterministic under seeded NumPy RNGs, writes per-step trajectories and summaries to `results/exhibit_6_1/`, computes η as the observed projected next state minus the harness model's projected next state, and asserts $\max |\eta| = 0$ in the main run. A TypeScript sketch can be added later if the exhibit is ported into a web-facing demo.

TRM/tinyLoRA extension

The FSM remains fixed; the occupant becomes a TRM router or small policy head. Per round, a tinyLoRA adapter is updated against the proxy exposed by the harness. Run both arms: private-objective drift and proxy-optimizing Goodhart. Measurement is unchanged: V from the FSM directly, no judge model, and $v \circ \phi$ asserted flat.

B Citation audit status

Verified enough for circulation, with visible scope

1. **Invariance entropy.** Colonius–Kawan 2009 and Kawan 2013 [?, ?] support the set-invariance/data-rate lineage. The linear or affine case is governed by unstable growth rates, while nonlinear volume-growth statements require regularity hypotheses such as admissible/control pairs, nonempty interior, positive measure, finite volume, or hyperbolic isolated-set assumptions. The manuscript therefore uses the linearized theorem as the load-bearing statement.
2. **Data-rate theorem.** The paper’s normalization is noiseless, quantized, and measured in bits per plant step. Nair–Evans [?] is the closest antecedent for that normalization. Tatikonda–Mitter and Sahai–Mitter are retained for the broader communication-constrained/noisy-channel setting, where channel hypotheses and reliability notions must be stated explicitly [?, ?].
3. **Viability theory.** Aubin [?] is used only for constrained-dynamics vocabulary: viable region, regulation map, and maintained constraint set. It is not cited for the entropy or width lower bound.

Downgraded or replaced

- Brockett 1997 is reinstated for the minimum-attention viewpoint [?]. Egerstedt–Brockett 2003 remains unpinned and is not used as a load-bearing citation. The draft also uses later, checkable sources for minimum-attention control and static coder-controller / lookup-table constructions [?, ?].
- Musat 2026 [?] is retained as preprint support for fixed-precision/description-length discipline and optional simplicity-prior motivation, not as support for the width theorem.

Still flagged

- The exact theorem/page references in Colonius–Kawan and Kawan should be page-verified before public release.
- Nair–Evans and Colonius–Kawan have stable volume/page metadata in the bibliography; exact theorem/page references should still be checked against the publisher records before public release. Tatikonda–Mitter DOI metadata remains optional because it is non-load-bearing here.
- Dean–Majumdar exact constants, including $\theta = (\ln 3)/4$, remain on the verification list; the manuscript uses only large-deviation suppression wording [?].
- Discrete-time Bode integral citations should be retained only if the fragility appendix is re-founded [?].
- Mori–Zwanzig and Koopman canonical picks remain non-load-bearing background [?, ?, ?].

C FRC as an Interactive Analogue of Time-Bounded Kolmogorov Complexity

Classical time-bounded Kolmogorov complexity asks for the shortest program that produces a fixed object within a bounded amount of computation:

$$K_U^t(y) = \min\{|p| : U(p) = y \text{ within time } t\}. \quad (35)$$

Fidelity Recursive Complexity (FRC) lifts this question from a static computation to an interactive, adversarial process. Rather than asking for the shortest program that computes an output once, it asks for the least defender resource budget that preserves a target property through repeated rounds of adaptive evaluation:

$$\text{FRC}_P(k; \delta) = \min\{R : \exists A \text{ such that } \text{Fid}_P^{(k)}(A) \geq 1 - \delta \text{ using resources } \leq R\}. \quad (36)$$

Here the fixed output y is replaced by a trajectory property P , the program p is replaced by an agent, controller, or harness A , the universal machine U is replaced by a rollout and evaluation process, and the time bound t is replaced by recursive depth k together with an explicit resource budget R . Informally: time-bounded Kolmogorov complexity prices computation; FRC prices maintained fidelity.

This analogy is intentionally imperfect. Kolmogorov complexity is a description-length quantity over programs, while FRC aggregates heterogeneous defender resources such as compute, data, oversight, proof effort, latency, attestation, and interface costs. The point of the analogy is not to identify FRC with Kolmogorov complexity, but to motivate the migration from producing an object once to keeping a property true under adaptive pressure.

D From FRC to Confinement Width

The earlier FRC formulation treated fidelity maintenance as a broad resource-minimization problem. For a property P , tolerance δ , and reflective depth k , it defined

$$\text{FRC}_P(k; \delta) = \min\{R : \exists A \text{ such that } \text{Fid}_P^{(k)}(A) \geq 1 - \delta \text{ using resources } \leq R\}. \quad (37)$$

This is a global adversarial property-testing quantity: it asks how much defender resource is required to preserve P against a k -round adaptive process that can condition on previous patches and failures.

The present note refactors that broad burden into a local dynamical object. Given an evaluator region

$$R_V = \{x : V(x) \geq \tau\}, \quad (38)$$

it isolates the interface width

$$W_V(R, T, \epsilon) = \min_{\theta} \min(s_r(\theta), s_w(\theta)), \quad (39)$$

the minimum number of quantized read/write channels required to keep trajectories inside R_V over the specified uncertainty class. Under the local hypotheses of Theorem 3.1, this width inherits a classical lower bound:

$$W(\theta) \geq \frac{1}{b} \sum_{|\lambda_i| > 1} \log_2 |\lambda_i|. \quad (40)$$

Thus the migration is from a global scalar resource burden to a roundwise decomposition:

$$\text{FRC}_V(K, \delta) \rightsquigarrow \inf_{\{\phi_k, G_k, \pi_k\}} \sum_{k=0}^{K-1} (W_{V,k} + A_k). \quad (41)$$

Here $W_{V,k}$ is the confinement-width cost of the round- k dynamics, while A_k is the audit and refinement cost of maintaining evaluator sufficiency and closure for the quarantine projection $\phi_k : X_k \rightarrow Z_k$. Confinement width is therefore a local lower-bound term inside FRC, while dimensional quarantine is the constructive route for reducing that term by forcing evaluator-relevant dynamics through a low-dimensional measured state.

In this sense, the older FRC program asked whether fidelity maintenance becomes intractable under recursive adaptation; the present note identifies one mathematically grounded summand of that cost.

References